Question			Answer	Marks	AOs	Guidance	
11	(a)		$\frac{dV}{dt} = \frac{k}{x}$	M1	3.3	Expresses inverse proportionality with a constant	
			When $t = 0,\ x = 5$ and $\frac{dV}{dt} = 21$, so $\frac{dV}{dt} = \frac{105}{x}$	A1	3.3	Evaluating k, oe (may be done later)	
			$\frac{dV}{dx} = 4\pi x^2$,	B1	2.1	may be embedded in chain rule	
			so the chain rule gives $\frac{dV}{dt} = \frac{dV}{dx} \times \frac{dx}{dt} = 4\pi x^2 \frac{dx}{dt}$	M1	2.1	Use of the chain rule	
			Hence $\frac{dx}{dt} = \frac{1}{4\pi x^2} \times \frac{105}{x} = \frac{105}{4\pi x^3}$ **AG**	A1 [5]	3.3	Convincing argument	
11	(b)		$\int 4\pi x^3 \, dx = \int 105 \, dt$	M1	3.1a	Separating the variables	
			$\pi x^4 = 105t + c$	A1	1.1b	Condone missing $+c$ here	
			When $t = 0, x = 5$ so $c = 625\pi$	M1 A1	3.3 3.3	Using initial conditions Correct value for c	
			When $t = 120$ $x = \sqrt[4]{\frac{105}{\pi} \times 120 + 625} = 8.25$ cm	A1 [5]	3.4	cao	
11	(c)		As t gets very large, the volume gets very large so the balloon will get beyond the maximum it can be without bursting and so burst.	E1 [1]	3.5b	Conveys the idea that $t \to \infty \Rightarrow V \to \infty$ or $x \to \infty$ Indicates a practical problem with very large volume	